

SET-BASED PROFILING

1. PROFILING

Profiling means studying the data **before and during data wrangling** to understand its **contents, quality, and behaviour**. Profiling is important because it guides the data transformations that may be required during data wrangling.

2. TYPES OF PROFILING

Individual Value Profiling	Set-Based Profiling
Individual value profiling determines whether individual record-field values are valid .	Set-based profiling does not concentrate on one value. It examines a set of values together .

3. INDIVIDUAL VALUE PROFILING

It determines whether individual record-field values are valid. It has two types:

- **Syntactic Profiling:** Syntax refers to the literal **format or form** of a value. Syntactic profiling checks whether a value belongs to the set of values permitted by the specified format or constraint.
- **Semantic Profiling:** Semantic profiling is concerned with the **meaning or interpretation** of a value.

4. SET-BASED PROFILING

A value can sometimes be **syntactically invalid but still have a meaningful interpretation in the context of the dataset**.

Set-based profiling shifts attention from an individual value to the **shape and extent of the distribution of values within a field**. It can also examine **relationships between multiple fields**.

What does set-based profiling examine?

Focus	What is examined?
Distribution	The shape and extent of the distribution of values within a field.
Multiple fields	Relationships between multiple fields.
Data quality	Patterns in the data that can help identify possible problematic values.

5. STATISTICS USED IN SET-BASED PROFILING

For **numeric data**, the following statistics are mentioned for examining the data:

Statistic	Meaning / Use
Minimum	The smallest value in the set.
Maximum	The largest value in the set.
Mean	The average value of the set.
Sum	The total of the values in the set.
Histogram	A representation used to examine the distribution of values.

6. WHY ARE THESE STATISTICS IMPORTANT?

Minimum, maximum, mean, sum, and histogram help us understand the **distribution of values** in the data. They can also help identify **possible outliers or problematic values**. Thus, set-based profiling gives a broader view of a field than checking individual values one by one.

QUICK REVISION

Individual Value Profiling → checks whether individual values are valid.

Set-Based Profiling → examines values as a set, their distribution, and relationships between fields.

Numeric statistics: MINIMUM → MAXIMUM → MEAN → SUM → HISTOGRAM